

SAMARTH DHOL

📞 123-456-7890 ✉️ sdhol@umass.edu 🔗 [linkedin.com/in/samarth-dhol-a68826258](https://www.linkedin.com/in/samarth-dhol-a68826258) 🐙 github.com/imsamarth123

Education

University of Massachusetts Amherst

Master of Science in Computer Science

Present

Amherst, MA

Charotar University of Science and Technology (CHARUSAT)

Bachelor of Technology in Computer Science & Engineering (CGPA: 9.58/10.0)

May 2026

Gujarat, India

Relevant Coursework

- Deep Learning
- Machine Learning
- Compiler Design
- Data Structures & Algorithms
- Large Language Models
- Cloud Computing
- Operating Systems

Experience

Indian Institute of Technology (IIT) Kharagpur

Dec 2025 – Mar 2026

Deep Learning Research Intern

West Bengal, India

- Designed and automated an end-to-end hybrid **deep learning pipeline** for optimal **3D sensor/camera placement** in indoor environments by reconstructing metric-scale 3D environments from RGB images to maximize visual coverage while satisfying physical mounting constraints.
- Developed a hybrid **3D reconstruction and optimization environment** by combining **COLMAP** and **DUST3R** for metric-scale scene reconstruction and integrating a custom **raycasting engine** for volumetric visibility and coverage evaluation.
- Solved the NP-hard **6-DoF sensor placement** problem using a hybrid **Recurrent PPO (LSTM)** and **CMA-ES** framework, improving volumetric coverage by **12.7%** and physical feasibility by **6.85%** over the baseline to achieve **82.57%** coverage and **91.76%** physical feasibility.

XYlofy AI

Apr 2026 – May 2026

AI & Data Analyst Intern

Remote

[\[GitHub Link\]](#) [\[Live Demo\]](#)

- Developed a **Breast Cancer Prediction** system using **Logistic Regression**, enabling automated classification of malignant and benign tumors to support preliminary clinical decision-making.
- Built an end-to-end machine learning pipeline involving data preprocessing, feature engineering, model training, and performance evaluation, achieving over **91% prediction accuracy** on a benchmark medical dataset.
- Engineered a secure **Django**-based web application with user authentication, integrating the trained model to deliver real-time breast cancer risk predictions through an intuitive interface for healthcare professionals.

Coding Blocks School Of Technology

Jun 2026 – Jul 2026

AI Engineer Intern

Remote

[\[GitHub Link\]](#) [\[Live Demo\]](#)

- Developed a **content-based movie recommendation system** using **Machine Learning** to generate personalized movie suggestions based on user preferences and movie metadata.
- Designed an end-to-end recommendation pipeline involving data preprocessing, feature extraction, similarity computation, and recommendation generation for real-time personalized predictions.
- Built and deployed an interactive **Streamlit** web application, enabling users to receive personalized movie recommendations through an intuitive interface.

Projects

Adaptive Robust Gradient Descent (ARGD): Dynamic Outlier-Resistant Regression

Python, NumPy, Scikit-learn, Pandas, Gradient Descent, Robust Optimization

Aug 2026

- Developed modified **Linear Regression** and **Multiple Linear Regression** algorithms featuring an adaptive gradient optimization strategy to improve robustness against outliers during model training.
- Implemented and benchmarked the proposed models against conventional regression algorithms on multiple benchmark datasets, including Salary, House Price Regression, Multiple Linear Regression, and California Housing datasets.
- Achieved up to a **17.4% improvement in R^2** over conventional Linear Regression on the Salary dataset (**0.6644** → **0.7803**) while consistently matching or outperforming standard regression baselines across multiple real-world benchmark datasets.

- Developed a Convolutional Neural Network (CNN) for automated pneumonia detection from chest X-ray images, achieving 99.5% classification accuracy on a public medical imaging dataset.
- Designed an end-to-end deep learning pipeline involving image preprocessing, model training, hyperparameter optimization, and performance evaluation for robust binary classification.
- Engineered an interactive Streamlit web application integrating the trained model to deliver real-time predictions with confidence scores, enabling accessible AI-assisted medical image analysis.

Publications

Yoga Pose Recognition Using MoveNet and Hybrid CNN-LSTM: A Deep Learning Approach for Posture Classification [\[Access Link\]](#)

ICDSA 2025

- Co-authored and published a conference paper presenting an AI-driven yoga pose recognition system using **MoveNet** and a hybrid **CNN-LSTM** architecture for posture classification.
- Designed and evaluated the deep learning pipeline, achieving **95.74% accuracy**, **95.33% F1-score**, and **99.32% AUC-ROC** on the Yoga Poses Dataset.

Honors & Awards

- **NPTEL Post Baccalaureate Fellowship - IIT Ropar:** Selected for a 1-year post-baccalaureate research fellowship under **Prof. Sukrit Gupta**, 2026.
- **NPTEL Deep Learning — Top 1% (Elite)**, scoring **92/100** in the certification examination, 2025.

Certifications

AWS Certified AI Practitioner

Jul 2025 – Jul 2028

Amazon Web Services — [\[Credential\]](#)

AWS Certified Cloud Practitioner

Jul 2024 – Jul 2027

Amazon Web Services — [\[Credential\]](#)

Technical Skills

Languages: Python, Java, C, SQL, JavaScript, HTML/CSS
Machine Learning / AI: Machine Learning, Deep Learning, Computer Vision, NLP, Reinforcement Learning
Frameworks / Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, Streamlit, Django
Tools / Platforms: Git, GitHub, AWS, COLMAP, DUST3R, Jupyter Notebook, VS Code